Weakly Paired Remote Sensing Change Captioning With Large Multimodal Model

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***Abstract*—Automatically generating textual descriptions of changes between bi-temporal remote sensing images remains challenging due to the labor-intensive and time-consuming process of obtaining fully aligned image–caption paired data. To address this limitation, we propose a novel weakly paired learning framework that harnesses the powerful visual and language capabilities of large multimodal models (LMMs) for remote sensing change captioning. Unlike conventional methods that rely on precisely annotated image-caption pairs, our approach effectively utilizes weakly paired tags that encapsulate changed objects and their corresponding actions for each image. The proposed framework is built upon an LMM architecture and operates synergistically across two distinct visual-language phases via parameter-efficient adaptation. In the visual change perception phase, we characterize spatiotemporal variations by identifying cross-temporal object changes and their associated actions through tag recognition. Subsequently, in the language interpretation phase, we leverage the pretrained LMM to convert these visual change tags into coherent and semantically rich textual descriptions, exploiting the linguistic knowledge embedded in the LMM. To further enhance the alignment of LMM-generated captions with human annotations, we introduce a train-time expansion strategy, which refines alignment by rewriting synonymously expressed captions back into their original phrasing. This strategy improves alignment without incurring additional inference costs. As a plug-and-play method, our framework is compatible with various LMMs and achieves performance comparable to state-of-the-art fully supervised approaches, demonstrating its effectiveness in remote sensing change captioning. The code will be available at https://github.com/mrazhou/WeCap**

***Index Terms*—Change captioning, large multimodal model (LMM), remote sensing, weakly paired.**

I. Introduction

THE rapid advancement of earth observation technologies has produced vast amounts of high-resolution remote sensing imagery, enabling detailed monitoring of land surface changes caused by human activities and natural hazards [1], [2], [3]. Remote sensing image change captioning (RSICC) has emerged as a key task to convert these bi-temporal image pairs into coherent, semantically accurate textual descriptions that explicitly answer *"what objects have changed and how"* [4], [5], [6]. By transforming complex visual changes into human-readable narratives, RSICC lowers the interpretation barrier for nonexperts and supports critical applications such as urban monitoring and disaster assessment.

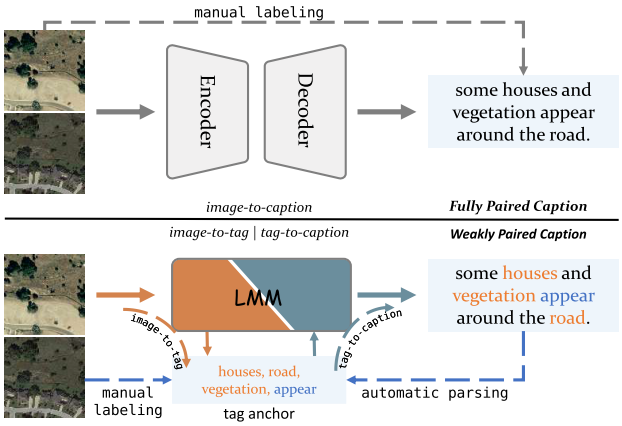


Fig. 1. (Top) Comparison of the conventional fully paired paradigm with (bottom) our factorized, weakly paired approach. The former requires costly end-to-end image-to-caption alignment, while WeCap introduces intermediate *tag anchors*—what changed and how it changed—to decouple the task into data-efficient image-to-tag perception and tag-to-caption interpretation stages.

As illustrated in Fig. 1, existing RSICC methods predominantly employ encoder–decoder architectures trained end-to-end on fully paired image–caption datasets. Early works enhanced change perception via contrastive pretraining [7], attentive fusion [5], or dual-branch transformers [8], [9]. Recent state-of-the-art (SOTA) approaches [10], [11], [12], [13] leverage pretrained large multimodal models (LMMs), integrating advanced visual encoders (e.g., SAM [14]) with powerful LLM decoders to improve localization and linguistic quality. However, all these methods rely heavily on expensive, expertly annotated image-caption pairs, severely limiting scalability. How to train RSICC models more data- and cost-efficiently remains a critical yet underexplored challenge.

To overcome this bottleneck, we propose weakly paired change captioning (WeCap), a novel framework that factorizes RSICC into two stages connected by concise visual tags representing the *"what"* (changed objects) and *"how"*

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(change actions). These tags act as semantic anchors, enabling weakly paired training that drastically reduces the need for full image–caption alignments. Tag-to-caption pairings can be automatically derived from existing captions via parsers, while image-to-tag data requires only lightweight labeling.

WeCap operates in two phases: 1) visual change perception (image-to-tag), distilling cross-temporal variations into structured tags and 2) language interpretation (tag-to-caption), leveraging pretrained LMMs to generate fluent descriptions. To improve alignment with human annotations, we introduce a train-time caption expansion strategy that rewrites synonymously expressed captions into canonical forms, enhancing linguistic consistency without inference overhead.

Compared to prior methods, WeCap delivers three key advantages: *Cost-Efficient Training*—substantially lower annotation burden via weakly paired tags; *Plug-and-Play Compatibility*—seamless integration with existing LMMs without architectural changes; and *Unilateral Scalability*—independent scaling of image-to-tag and tag-to-caption data for flexible, low-cost performance gains.

Our main contributions are as follows.

1) The first weakly paired learning framework for RSICC. By introducing visual tags as intermediate semantic anchors, WeCap greatly reduces reliance on costly fully paired image–caption data while preserving high performance.
2) A train-time caption expansion strategy that refines linguistic consistency by normalizing synonymous expressions into canonical phrasings, boosting fidelity to human annotations at no extra inference cost.
3) Comprehensive experiments across seven LMM backbones, demonstrating WeCap's superior effectiveness, transparency, compatibility, and scalability—achieving performance on par with fully supervised SOTA while dramatically cutting annotation costs.

II. Related Work

A. Remote Sensing Image Change Captioning

The dominant paradigm in RSICC is the encoder–decoder framework [5], [7], [9], [15]. Early methods [5], [8] used Siamese CNNs to extract features from bi-temporal images, followed by differencing or concatenation to obtain change representations. More advanced approaches employ cross-attention [4], [7] or transformer-based encoders [6] to capture fine-grained interimage interactions. The decoder, typically an RNN [16] or Transformer [7], [8], [17], then autoregressively generates captions from these features.

Recent works shift toward pretrained LMMs, adapting them via specialized fine-tuning. Examples include Semantic-CC [10], which integrates SAM with MiniGPT4 using pixel-level semantic guidance, and instruction-tuned models like ChangeChat [11] and KCFI [12], which incorporate visual instructions and auxiliary change detection tasks to focus on relevant changes. More recently, new large-scale benchmarks such as MOSAIC-SEN2-CC [18] and SECOND-CC [19] have been introduced to address multispectral data modeling and robustness in complex environments. These works provide valuable insights into dataset diversity and robust feature extraction.

Despite these advances, nearly all existing RSICC methods rely on fully supervised training with large-scale datasets of bi-temporal image pairs paired with ground-truth captions ($\mathcal{D}_{\text{full}}$). Creating such datasets is labor-intensive and requires domain expertise, posing a major barrier to scalability. Our work addresses this limitation by introducing a factorized, weakly paired paradigm that eliminates the need for costly fully paired annotations.

B. Large Multimodal Models in Vision and Language

The advent of LMMs, such as Qwen2-VL [20], LLaVA [21], and InternVL [22], has revolutionized the field of vision-language research. By aligning powerful pretrained vision encoders with LLMs, these models have demonstrated remarkable capabilities in few-shot and zero-shot learning, instruction following, and complex reasoning across modalities.

In the domain of remote sensing, researchers have begun to explore the potential of LMMs for various tasks, including visual question answering [23], [24], object detection [25], [26], and image classification [27], [28]. These works have shown that fine-tuning LMMs on remote sensing data can adapt their vast world knowledge to this specialized domain. However, most of these adaptation efforts still rely on fully annotated datasets. Our work diverges from this trend by not only leveraging the powerful architecture of an LMM but also by proposing a novel training strategy that significantly reduces the annotation requirements, making the adaptation of LMMs to RSICC more data-efficient.

C. Weakly Supervised Image Captioning

The high cost of paired image-caption data is a common challenge in image captioning, prompting extensive research into weakly supervised methods that learn image-to-text mappings using weaker supervision signals.

One key direction uses image-level labels (e.g., object tags) instead of full captions. Approaches include graph neural networks to model object relationships without bounding boxes [29] and grounded captioning frameworks that enhance localization and spatial context [30], [31], [32]. Another major line explores semi-supervised and unpaired learning. Effective techniques include teacher–student frameworks with pseudo-labeling and noise injection [33], adversarial training via GANs for semantic consistency [34], [35], and cross-modal regularizers to maintain visual-textual alignment [36], [37].

Despite success in general-domain captioning, these methods remain largely unexplored for the more complex task of RSICC. Our work introduces the first principled weakly paired framework for RSICC, uniquely combining semantic tags as intermediate supervision with the generative capabilities of modern LMMs, while proposing a novel Linguistic Alignment self-supervision task to address linguistic variations.

III. Methodology

As shown in Fig. 2, our methodology introduces a paradigm shift in RSICC, moving from a monolithic data dependency to a decomposed, weakly paired framework. We first

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formalize this new paradigm, highlighting its fundamental statistical advantages over conventional approaches (Section III-A). We then detail the two core capabilities of our WeCap framework—visual change perception (Section III-B) and language interpretation (Section III-C)—which are learned concurrently within a single, unified model. Subsequently, we present our novel training strategy for aligning the model's generative space with the rich manifold of human language (Section III-D). Finally, we formalize the composed inference pipeline that seamlessly executes the model's trained capabilities (Section III-E).

A. Weakly Paired Paradigm for RSICC

Let $\mathcal{P}$, $\mathcal{T}$, and $\mathcal{C}$ denote the spaces of image pairs, semantic tag sets, and natural language captions, respectively. The ultimate goal of RSICC is to model the conditional probability distribution $p(C|P)$, where $P \in \mathcal{P}$ and $C \in \mathcal{C}$.

1) Conventional Monolithic Paradigm: Existing methods implicitly or explicitly model $p(C|P)$ through a single, monolithic function $f_\theta : \mathcal{P} \to \mathcal{C}$. The optimization of parameters θ is contingent upon a large dataset $\mathcal{D}_{\text{full}}$ containing samples drawn from the true joint distribution $p(P, C)$. The learning objective is thus

$$\theta^* = \arg\max_\theta \mathbb{E}_{(P,C)\sim p(P,C)}[\log p(C|P;\theta)]. \tag{1}$$

This paradigm's critical limitation is its reliance on sampling $p(P, C)$, a process that is notoriously expensive and constitutes the primary bottleneck for scalability.

2) Our Factorized Weakly Paired Paradigm: We propose to circumvent the need for the joint distribution $p(P, C)$ by introducing an intermediate variable, the semantic tag set $T \in \mathcal{T}$. By the chain rule of probability and a key semantic sufficiency assumption—that the tags T capture sufficient information from the image pair P for generating the caption C, i.e., $p(C|T, P) \approx p(C|T)$—we decompose the problem into learning two separate conditional distributions: $p(T|P)$ and $p(C|T)$. These distributions can be learned from two independent and more economical datasets, $\mathcal{D}_{P\to T} \sim p(P, T)$ and $\mathcal{D}_{T\to C} \sim p(T, C)$. This factorization allows us to conceptualize the problem as learning two distinct mappings, both of which are materialized by a single unified model.

1) *Perception:* A mapping from $\mathcal{P}$ to $\mathcal{T}$ that models $p(T|P)$.
2) *Interpretation:* A mapping from $\mathcal{T}$ to $\mathcal{C}$ that models $p(C|T)$.

This factorization replaces a single, hard learning problem with two simpler, more data-efficient ones.

B. Prompt-Conditioned Visual Perception

The first capability our unified model must learn is perception: modeling the distribution $p(T|P)$. We adapt a foundational LMM, $\mathcal{M}_\Theta$, using a single set of parameter-efficient fine-tuning (PEFT) parameters, denoted ϕ, which are shared across all tasks. When presented with a sample from the image-to-tag dataset $\mathcal{D}_{P\to T}$, the model is conditioned using an image pair $P = (I_{\text{pre}}, I_{\text{post}})$ and a specific instructional prompt, Π_{perceive}:

> "Please identify the objects involved in the change in the two pictures and the behavior of the change."

This prompt cues the model to enter its "perception mode." The fine-tuned model, $\mathcal{M}_{\Theta,\phi}$, then performs a conditional generative process, autoregressively outputting a sequence of semantic tags $T = (t_1, \ldots, t_k)$. The probability of generating a specific tag sequence is given by

$$p(T|P;\phi) = \prod_{i=1}^{k} p\left(t_i|t_{<i}, P, \Pi_{\text{perceive}}; \mathcal{M}_{\Theta,\phi}\right). \tag{2}$$

The target output for this task is a sequence of words, and the model is trained to maximize the likelihood of the ground-truth tag sequence using a standard cross-entropy loss.

C. Prompt-Conditioned Language Interpretation

The second capability is interpretation: modeling the distribution $p(C|T)$. The same LMM with the same PEFT parameters ϕ is used. The model seamlessly switches its functionality when it detects an input formatted for the interpretation task, i.e., a sample from the tag-to-caption dataset $\mathcal{D}_{T\to C}$. For such samples, the input tags are programmatically structured into a different prompt, $\Pi_{\text{interpret}}(T)$:

> "Please generate a sentence to describe the difference with the objects and actions: Objects: $\{t_{\text{obj}_1}, \ldots,\}$, Actions: $\{t_{\text{act}_1}, \ldots,\}$"

Conditioned on this purely textual, structured prompt, the model $\mathcal{M}_{\Theta,\phi}$ enters its "interpretation mode." It then autoregressively generates a full, natural language caption $C = (c_1, \ldots, c_m)$, modeling the probability

$$p(C|T;\phi) = \prod_{j=1}^{m} p\left(c_j|c_{<j}, \Pi_{\text{interpret}}(T); \mathcal{M}_{\Theta,\phi}\right). \tag{3}$$

This task leverages the same cross-entropy loss objective to maximize the likelihood of the ground-truth caption.

D. Dual-Component Linguistic Alignment

To explicitly enhance linguistic robustness, we introduce a third training task: linguistic alignment. This task is composed of two complementary subtasks that collectively regularize the model's language space.

1) Denoising Alignment: This subtask trains the model to map diverse, synonymously expressed sentences back to their canonical form. We define a stochastic transformation operator $\mathcal{A} : \mathcal{C} \to \mathcal{C}$ that generates a paraphrase $C' = \mathcal{A}(C, p_{replace})$ of a caption C via random synonym replacement with probability $p_{replace}$.[1] The model is conditioned with a denoising prompt, $\Pi_{\text{denoise}}(C')$:

> "Write the original sentence according to the synonymous sentence expressions provided: $\{C'\}$"

It is then tasked with regenerating the original sentence C, thereby learning the denoising distribution $p(C|C';\phi)$. This forces the model to recognize semantic equivalence across different surface forms.

[1]Our implementation of the operator $\mathcal{A}$ leverages the NLTK library for part-of-speech (POS) tagging and the WordNet corpus for retrieving valid synonyms.

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2) Identity Alignment: To anchor the model and prevent it from over-correcting already well-formed sentences, we add an identity mapping subtask. The model is given a clean sentence C and tasked with reproducing it exactly. This is guided by an identity prompt, $\Pi_{\text{identity}}(C)$:

"Output the given sentence: {C}"

This task reinforces the stability of canonical sentence structures within the model's learned representations.

E. Unified Training and Composed Inference

The core of our training strategy is its unification. We construct a single heterogeneous training dataset by mixing samples from all three tasks. Let $\mathcal{D}_{P\to T}$ and $\mathcal{D}_{T\to C}$ be the primary datasets. We derive a third dataset, $\mathcal{D}_{\text{align}} = \{(\mathcal{A}(C), C)\}_{\text{Denoising}} \cup \{(C, C)\}_{\text{Identity}} \quad \forall (T, C) \in \mathcal{D}_{T\to C}$. The unified training set is $\mathcal{D}_{\text{unified}} = \mathcal{D}_{P\to T} \cup \mathcal{D}_{T\to C} \cup \mathcal{D}_{\text{align}}$.

The model is fine-tuned on this mixed dataset using a single, task-agnostic loss function: the standard cross-entropy loss for sequence generation. Let a generic training sample be (x, y), where x is the conditioning input and y is the target sequence. The optimization objective is to find the parameters ϕ^* that minimize the negative log-likelihood over the unified dataset

$$\phi^* = \arg\min_{\phi} \mathbb{E}_{(x,y)\sim\mathcal{D}_{\text{unified}}} \left[-\log p(y|x;\phi) \right]. \tag{4}$$

The model processes each sample based on its structure.

1) If (x, y) is a perception sample, then $x = (P, \Pi_{\text{perceive}})$ and $y = T$.
2) If (x, y) is an interpretation sample, then $x = \Pi_{\text{interpret}}(T)$ and $y = C$.
3) If (x, y) is for denoising alignment (DA), then $x = \Pi_{\text{denoise}}(C')$ and $y = C$.
4) If (x, y) is for identity alignment (IA), then $x = \Pi_{\text{identity}}(C)$ and $y = C$.

This elegant formulation allows the single set of parameters ϕ to simultaneously learn all three complementary capabilities.

Inference, notably, remains a two-stage pipeline that does not involve the alignment task. For a new image pair P, the fully trained model $\mathcal{M}_{\Theta,\phi^*}$ first executes the Perception task to generate tags $\hat{T}$, then immediately executes the Interpretation task on $\hat{T}$ to produce the final caption $\hat{C}$. This demonstrates how a single, versatile model, trained via our factorized, weakly paired paradigm, can efficiently support the full RSICC pipeline.

IV. Experiments

In this section, we conduct a comprehensive set of experiments to validate the effectiveness of our proposed WeCap framework. We aim to answer the following key questions: 1) how does WeCap, under a weakly paired setting, perform against SOTA fully supervised methods? 2) can WeCap effectively leverage additional, unpaired data to boost performance? 3) how well does the learned model generalize to unseen datasets? and 4) what is the contribution of each component in our unified training strategy?

TABLE I

Performance Comparisons on LEVIR-CC. The Best Results Among Existing Fully Paired Methods are Colored in Blue. The Best Results Achieved by Our WeCap Framework are Highlighted in Red. Within the Existing Methods, the Dashed Line Separates Traditional Encoder–Decoder Architectures From More Recent Approaches Based on LMMs

Method	B-1	B-2	B-3	B-4	M	R-L	C	S_m^*
Existing Methods (Fully Paired)								
MCCFormer-S [38]	79.90	70.26	62.68	56.68	36.17	69.46	120.39	70.68
MCCFormer-D [38]	80.42	70.87	62.86	56.38	37.29	70.32	124.44	72.11
ICT-Net [39]	84.45	75.69	67.94	61.54	38.83	73.25	130.23	75.96
PSNet [9]	83.86	75.13	67.89	62.11	38.80	73.60	132.62	76.78
RSICCformer [8]	84.72	76.27	68.87	62.77	39.61	74.12	134.12	77.65
PromptCC [6]	83.66	75.73	68.10	63.54	38.82	73.72	136.44	78.13
Pix4Cap [40]	85.56	77.08	69.79	63.78	39.96	75.12	136.76	78.91
Chg2Cap [5]	86.14	78.08	70.66	64.39	40.03	75.12	136.61	79.04
SFT [41]	84.40	76.00	68.24	61.84	39.88	74.57	135.03	77.83
SEN [7]	85.10	77.05	70.01	64.09	39.59	74.57	136.02	78.57
RSCaMa [42]	85.79	77.99	71.04	65.24	39.91	75.24	136.56	79.24
LLaVA-1.5 [21]	-	-	-	-	-	-	126.25	-
GeoChat [27]	-	-	-	-	-	-	128.36	-
UniRS [43]	-	-	-	-	-	-	139.12	-
Qwen2-VL-7B [20]	84.46	76.04	69.03	63.30	39.68	73.89	134.07	77.73
InternVL2-8B [22]	82.73	72.79	64.32	57.27	38.37	72.86	131.19	74.92
LLaVA1.5-7B [21]	83.81	74.25	66.49	60.03	40.09	73.99	135.96	77.52
ChangeMinds [44]	83.14	-	-	-	38.73	74.01	136.56	-
KCFI [12]	86.34	77.31	70.89	65.30	39.42	75.47	138.25	79.61
Weakly Paired Training (Ours)								
WeCap (Ours)								
↪ *w/ Qwen2-VL-2B*	84.63	75.81	68.42	62.56	39.39	73.64	133.02	77.15
↪ *w/ Qwen2-VL-7B*	85.19	76.21	68.76	62.88	40.15	74.92	136.47	78.60
↪ *w/ InternVL2-2B*	80.66	70.00	61.01	53.46	37.85	72.42	130.08	73.45
↪ *w/ InternVL2-8B*	84.80	74.92	66.95	60.75	39.00	73.21	133.33	76.57
↪ *w/ LLaVA1.5-7B*	84.56	76.25	69.13	63.63	40.04	74.52	135.87	78.52

A. Datasets and Experimental Settings

1) Datasets: Our experiments are conducted on two widely used public datasets for RSICC.

1) *LEVIR-CC [8]:* A large-scale benchmark derived from LEVIR-CD, consisting of 10 077 bi-temporal 256 × 256 image pairs (0.5 m/pixel) from 20 regions, spanning 5–15 years. Balanced in changed/unchanged scenes, it focuses on urban development changes. Each pair has five descriptive captions (over 50 000 sentences total). The full vocabulary is retained to preserve semantic integrity and evaluate handling of long-tail word distributions.
2) *WHU-CDC [4]:* A diverse dataset with 7434 bi-temporal 256 × 256 image pairs at higher resolution (0.075 m/pixel), capturing post-earthquake reconstruction in Christchurch, New Zealand. Changes include buildings, roads, and vegetation. Each pair is annotated with five expert-provided captions (over 37 000 sentences total).

To simulate the weakly paired setting, we automatically process the ground-truth captions of the training sets to create

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TABLE II

PERFORMANCE COMPARISONS ON WHU-CDC. THE BEST RESULTS AMONG EXISTING FULLY PAIRED METHODS ARE IN BLUE. THE BEST RESULTS ACHIEVED BY OUR WECAP FRAMEWORK ARE HIGHLIGHTED IN RED

Method	B-1	B-2	B-3	B-4	M	R-L	C	S_m^*
Existing Methods (Fully Paired)								
DUDA [16]	79.04	69.53	61.57	55.64	34.29	68.98	121.85	70.19
MCCFormer-S [38]	82.14	76.29	71.08	66.51	43.50	79.76	148.88	84.66
MCCFormer-D [38]	73.29	67.88	64.03	60.96	39.69	73.67	134.92	77.31
RSICCformer-C [8]	78.25	72.82	68.57	65.14	44.35	76.50	143.44	82.34
RSICCformer [8]	80.05	74.24	69.61	66.54	42.65	73.91	133.44	79.14
CTMTNet [4]	83.56	77.66	72.76	69.00	45.39	79.23	149.40	85.76
Chg2Cap [5]	86.14	80.89	76.99	74.31	46.96	80.27	151.15	88.17
SEN [7]	86.21	81.27	77.49	74.78	47.85	81.57	155.11	89.83
Qwen2-VL [20]	83.78	77.89	72.82	68.59	46.30	80.84	154.01	87.44
Weakly Paired Training (Ours)								
WeCap (Ours)								
↪ *w/ Qwen2-VL-2B*	84.48	77.76	73.30	69.98	45.68	80.44	153.33	87.36
↪ *w/ Qwen2-VL-7B*	86.08	80.85	77.04	74.56	47.00	80.58	152.27	88.60
↪ *w/ InternVL2-2B*	84.21	78.15	73.01	68.56	46.82	81.39	157.76	88.63
↪ *w/ InternVL2-8B*	87.25	81.88	77.84	75.12	47.95	81.65	154.98	89.93
↪ *w/ LLaVA1.5-7B*	85.50	80.50	77.16	75.01	47.19	80.61	152.39	88.80

TABLE III

DEMONSTRATION OF WECAP'S UNILATERAL DATA SCALING. OUR FRAMEWORK, SEEDED WITH ONLY 10% OF THE IMAGE-TAG DATA ($\mathcal{D}_{P\to T}$) AND TAG-CAPTION DATA ($\mathcal{D}_{T\to C}$), CAN LEVERAGE THE FULL 100% OF EITHER COMPONENT TO SIGNIFICANTLY BOOST PERFORMANCE. BASELINES ARE TRAINED ON THE EQUIVALENT 10% FULLY PAIRED DATA. ALL WECAP VARIANTS INCLUDE THE CORRESPONDING ALIGNMENT DATA ($\mathcal{D}_{\text{ALIGN}}$) DERIVED FROM THEIR $\mathcal{D}_{T\to C}$ COMPONENT. THE BEST RESULT PER METRIC IS IN BOLD

Method	B-1	B-2	B-3	B-4	M	R-L	C	S_m^*
LEVIR-CC Dataset								
SEN	78.20	70.25	63.73	58.34	35.03	69.49	119.22	70.52
Qwen2-VL-2B SFT	79.44	69.28	61.20	54.44	36.58	70.66	125.66	71.84
WeCap (10%)	80.77	71.65	63.77	57.76	37.28	70.56	124.52	72.53
+ *100%* $\mathcal{D}_{T\to C}$	81.94	71.58	64.15	58.90	37.31	71.65	128.06	73.98
+ *100%* $\mathcal{D}_{P\to T}$	**83.00**	**72.47**	**63.79**	**56.78**	**37.95**	**72.42**	**131.78**	**74.73**
WHU-CDC Dataset								
SEN	84.08	78.69	74.70	72.15	45.55	78.42	145.37	85.37
Qwen2-VL-2B SFT	72.91	64.93	58.90	53.88	38.06	70.25	123.19	71.34
WeCap (10%)	82.07	75.33	70.03	65.54	43.11	79.64	152.39	85.17
+ *100%* $\mathcal{D}_{T\to C}$	84.46	**79.39**	**75.71**	**73.27**	**46.67**	**79.94**	151.24	**87.78**
+ *100%* $\mathcal{D}_{P\to T}$	**84.67**	78.75	74.03	70.68	46.58	79.87	**152.48**	87.40

our factorized datasets. For each fully paired sample (P, C), we use sentence parsers to extract a set of keyword tags T, thereby generating an image-to-tag sample (P, T) for $\mathcal{D}_{P\to T}$ and a tag-to-caption sample (T, C) for $\mathcal{D}_{T\to C}$.

2) Evaluation Metrics: Following standard practice in image captioning, we evaluate performance using a comprehensive suite of metrics: BLEU (B-1–B-4) [45], METEOR (M) [46], ROUGE-L (R-L) [47], and CIDEr (C) [48]. These metrics assess various aspects of caption quality, including n-gram precision, recall, alignment, and consensus with reference captions. Additionally, we report the summary score S_m^*, the arithmetic mean of the four principal metrics: B-4, M, R-L, and C, providing a holistic measure of overall performance.

3) Implementation Details: Our WeCap framework is implemented on top of several publicly available LMMs, including the Qwen2-VL [20], InternVL2 [22], and LLaVA1.5 [21] series, to demonstrate its plug-and-play compatibility. We employ PEFT using low-rank adaptation (LoRA) [49] with a rank of $r = 8$ and an alpha of $\alpha = 32$. The models are trained on a unified, mixed dataset as described in our methodology. We use the AdamW optimizer with a learning rate of 1e-4. To accommodate GPU memory constraints, the batch size is set to 8 for 2 B models and 1 for the larger 7/8 B models. All experiments are conducted on four NVIDIA RTX 4090 GPUs. For the alignment task, the synonym replacement probability is set to $p_{replace} = 0.6$, based on our ablation study.

TABLE IV

EVALUATION OF CROSS-DATASET GENERALIZATION. MODELS ARE TRAINED ON A SOURCE DOMAIN ($\mathcal{S}$) AND DIRECTLY EVALUATED ON AN UNSEEN TARGET DOMAIN ($\mathcal{T}$). WECAP IS BUILT UPON THE QWEN2-VL-2B BACKBONE

Method	B-1	B-2	B-3	B-4	M	R-L	C	S_m^*
Transfer: LEVIR ($\mathcal{S}$) → WHU ($\mathcal{T}$)								
SEN	52.20	41.68	37.73	35.47	21.57	44.69	66.39	42.03
Qwen2-VL-2B	79.10	72.21	67.38	63.71	44.78	77.22	142.97	82.17
WeCap	**83.31**	**77.07**	**72.71**	**69.61**	**46.21**	**79.45**	**149.48**	**86.19**
Transfer: WHU ($\mathcal{S}$) → LEVIR ($\mathcal{T}$)								
SEN	59.08	49.39	45.47	43.49	22.60	51.41	81.93	49.86
Qwen2-VL-2B	72.70	61.92	54.10	48.63	32.28	64.20	101.76	61.72
WeCap	**72.91**	**62.86**	**56.07**	**51.39**	**32.53**	**65.77**	**112.02**	**65.43**

B. Comparison With SOTA Methods

We first compare WeCap against existing SOTA methods on the LEVIR-CC and WHU-CDC datasets. The results are presented in Tables I and II, respectively. For a fair comparison, all existing methods are trained on the full 100% paired dataset, while our WeCap is trained under the weakly paired setting described in our methodology.

1) Performance Analysis: The results compellingly demonstrate that our data-efficient WeCap framework is not only competitive but can also outperform fully supervised SOTA methods. On the LEVIR-CC dataset (Table I), WeCap showcases remarkable competitiveness. For instance, when equipped with the LLaVA1.5-7B backbone, its BLEU-4 score of 63.63 closely approaches the fully supervised SOTA of 65.30 achieved by KCFI. This is a significant accomplishment, highlighting our method's ability to nearly match the performance of fully paired approaches despite its significantly lower supervision requirements.

This advantage becomes even more pronounced on the more challenging WHU-CDC dataset (Table II). Our WeCap framework, when integrated with the InternVL2-8B model,

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TABLE V

ABLATION STUDIES ON THE COMPONENTS OF OUR WECAP FRAMEWORK, USING QWEN2-VL-2B AS THE BASE MODEL. P, I, DA, AND IA REPRESENT THE PERCEPTION, INTERPRETATION, DA, AND IA TASKS, RESPECTIVELY. OUR FULL MODEL DEMONSTRATES CONSISTENT IMPROVEMENTS ACROSS ALL METRICS

Method	P	I	DA	IA	B-1	B-2	B-3	B-4	M	R-L	C	S_m^*
Full Paired SFT	-	-	-	-	82.52	73.44	66.06	60.28	39.35	73.51	130.94	76.02
WeCap (Base)	✓	✓			80.68	69.66	59.45	50.39	35.57	72.12	130.82	72.22
WeCap (+DA)	✓	✓	✓		83.31	73.77	65.97	60.02	38.87	72.77	131.03	75.67
WeCap (+IA)	✓	✓		✓	83.35	73.74	65.73	59.31	38.95	72.92	131.06	75.56
WeCap (Full)	✓	✓	✓	✓	**84.63**	**75.81**	**68.42**	**62.56**	**39.39**	**73.64**	**133.02**	**77.15**

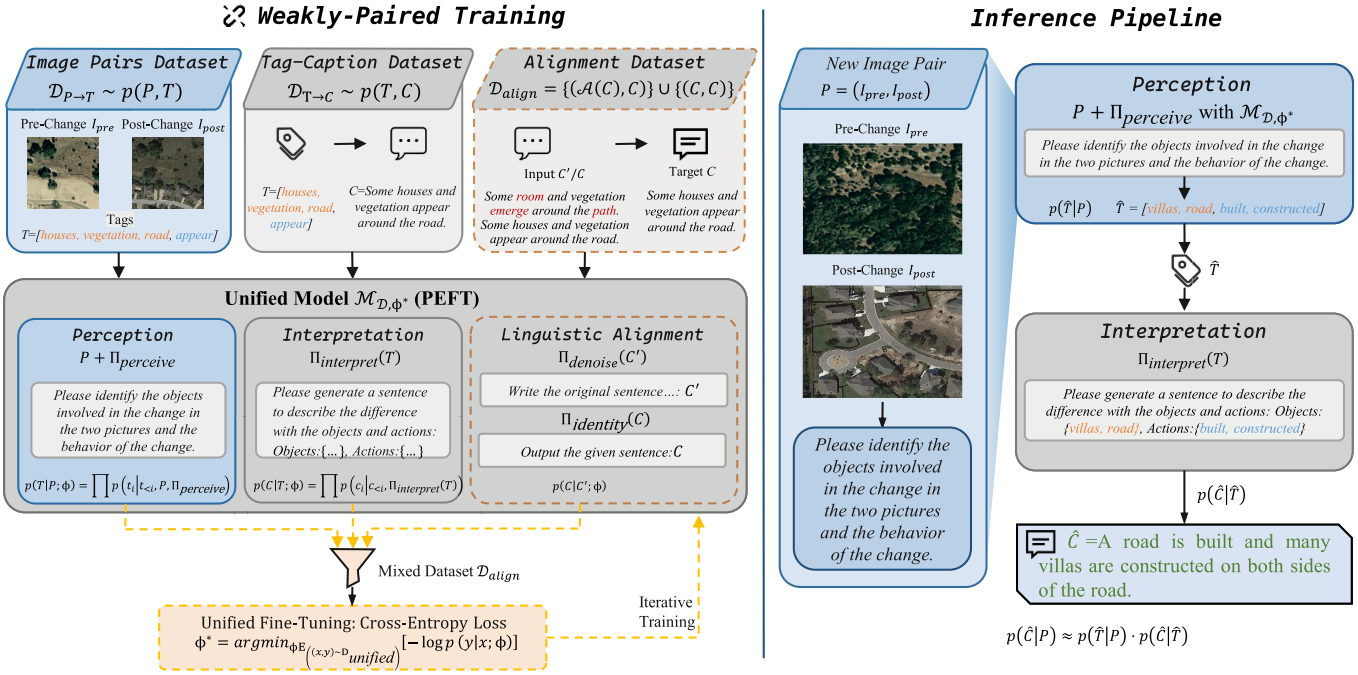


Fig. 2. Overview of our WeCap framework's training and inference process. Our approach leverages a single LMM, fine-tuned on a mixture of three weakly paired data sources to learn three distinct capabilities: perception (image pair → tags), interpretation (tags → caption), and linguistic alignment (paraphrase → canonical caption). Each capability is invoked by a specific prompt. During inference, the trained model operates in a seamless two-stage pipeline: it first *perceives* a set of semantic tags from a new image pair, and then *interprets* these tags to generate the final, descriptive caption.

establishes a new SOTA. It outperforms the formidable SEN baseline across nearly every metric and sets a new record for the summary score S_m^* (89.93 versus 89.83). These results are remarkable, as they show that by factorizing the learning problem and using a more effective training strategy, our weakly paired method can achieve and even surpass the performance of monolithic models that rely on expensive, fully aligned data.

2) Plug-and-Play Compatibility: A key strength of WeCap is its model-agnostic, plug-and-play nature. As shown in both tables, we have successfully applied our framework to five different LMMs from three distinct model families (Qwen, InternVL, and LLaVA) with varying parameter counts (2–8 B). In all cases, the WeCap framework successfully adapts the base LMM to the RSICC task, and performance scales gracefully with the model's inherent capabilities. This demonstrates that WeCap is not a specialized solution tailored to a single architecture but a general and robust framework for weakly paired change captioning.

3) Relative Performance Improvements Across Backbones: Beyond absolute metrics, WeCap consistently provides substantial relative improvements over the fully paired fine-tuning (SFT) of the respective base models. For instance, while the InternVL2-8B SFT baseline achieves an S_m^* of 74.92, applying WeCap boosts it to 76.57. Similarly, LLaVA1.5-7B improves from 77.52 (SFT) to 78.52 (WeCap). This consistent enhancement across diverse LMM architectures (including Qwen, InternVL, and LLaVA) strongly validates that the performance gains stem from our proposed decoupled, weakly paired paradigm rather than merely relying on the strong priors of the underlying backbones.

C. Unilateral Data Scalability

This experiment validates the core hypothesis of this article: that the factorized paradigm allows for flexible and efficient data scaling. We seed our WeCap framework with only 10% of the available image-tag data ($\mathcal{D}_{P\to T}$) and tag-caption data ($\mathcal{D}_{T\to C}$), and then unilaterally scale one of these components

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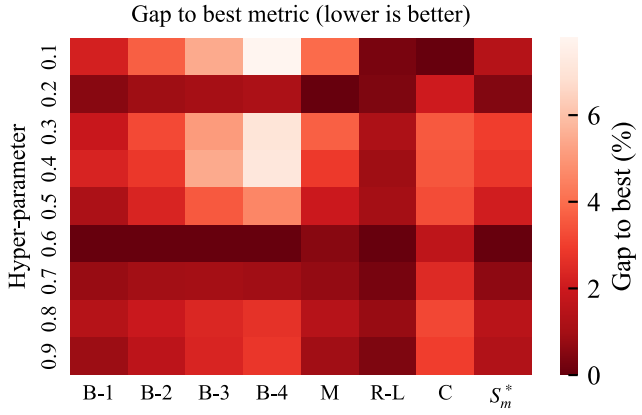


Fig. 3. Sensitivity analysis of the synonym replacement probability, $p_{replace}$, used in our linguistic alignment task. The heatmap visualizes the percentage gap to the best-achieved score for each metric across different probability values. Darker red indicates a smaller gap and thus better performance.

TABLE VI
ABLATION STUDY ON TAG NOISE ROBUSTNESS DURING THE INTERPRETATION PHASE

Tag Quality Setting	B-1	B-4	M	C	S_m^*
WeCap (Base)	84.63	62.56	39.39	133.02	77.15
Drop 60% Tags	80.89	58.77	36.94	127.73	73.81
Drop 80% Tags	73.16	54.02	32.47	114.66	66.83
Without Tags	59.99	47.64	26.81	101.63	58.95
Add 1 Noise Tag	58.41	32.14	30.13	33.45	35.32

TABLE VII
COMPARISON OF FINE-GRAINED VOCABULARY RECALL ON THE LEVIR-CC TEST SET

Method	Spatial Recall	Quantity Recall
SEN (Fully Paired)	15.70%	**30.70%**
Qwen2-VL-7B SFT (Fully Paired)	14.28%	25.43%
WeCap (Weakly Paired)	**21.65%**	25.60%

to 100% while keeping the other at 10%. The results are presented in Table III.

The findings are unequivocal. Starting from the 10% baseline, we observe substantial performance improvements by adding more of either data type. On LEVIR-CC, scaling the image-tag dataset to 100% boosts the S_m^* score from 72.53 to 74.73. On WHU-CDC, scaling the tag-caption dataset to 100% yields a remarkable jump in S_m^* from 85.17 to 87.78, surpassing many fully supervised baselines. This powerfully demonstrates that our WeCap framework effectively breaks the data bottleneck of traditional methods. It can efficiently leverage cheaper, weakly paired data sources to enhance performance, a capability that monolithic, fully paired models inherently lack.

D. Cross-Dataset Generalization

A robust model should generalize well to unseen data distributions. We evaluate this by training models on one dataset (source, $\mathcal{S}$) and directly testing them on another (target, $\mathcal{T}$) without any fine-tuning. As shown in Table IV, traditional fully supervised methods like SEN suffer a catastrophic performance degradation, indicating severe overfitting to the source domain's specific biases.

While the fully supervised LMM (Qwen2-VL-2B) shows better generalization, our WeCap framework demonstrates a markedly superior transferability in both directions. For instance, when transferring from LEVIR to WHU, WeCap achieves an S_m^* score of 86.19, a massive improvement over both SEN (42.03) and the fully paired Qwen model (82.17). This suggests that by learning from disentangled, weaker signals, WeCap acquires more fundamental and transferable representations of visual change, making it more robust and reliable for real-world applications.

TABLE VIII
EVALUATION OF GENERAL CAPABILITIES TO ANALYZE VOCABULARY BIAS

Model	MMMU	MME	AI2D	MMStar	ChartQA
Qwen2-VL-2B (Base)	40.56	376.07	69.72	42.64	70.92
WeCap-2B (Ours)	40.09	377.50	69.02	41.97	70.09

E. Ablation Studies

We conduct ablation studies on LEVIR-CC with Qwen2-VL-2B to dissect the contributions of our proposed training components and analyze the key hyperparameters.

1) Effectiveness of Alignment Tasks: To validate the importance of our Linguistic Alignment tasks, we start with a base WeCap model trained only on Perception and Interpretation (P+I) and progressively add the DA and IA tasks. As shown in Table V, the base model already provides a solid foundation. The introduction of either DA or IA alone yields significant performance gains, with DA being particularly effective. Crucially, the full WeCap model, which incorporates all four training tasks (P+I+DA+IA), achieves the best results across all metrics. This confirms that the two alignment tasks are complementary and act as effective regularizers, enhancing the model's linguistic robustness and overall performance.

2) Analysis of Synonym Replacement Probability: The Linguistic Alignment task is governed by the synonym replacement probability, $p_{replace}$. To find its optimal value, we vary it from 0.1 to 0.9. The results are visualized in Fig. 3. We observe a clear "sweet spot," where a replacement probability of $p_{replace} = 0.6$ achieves the most consistent, near-optimal performance across the majority of metrics. Excessively low probabilities fail to introduce enough linguistic diversity, while excessively high probabilities can corrupt the sentence structure. Based on this analysis, we use $p_{replace} = 0.6$ for all other experiments.

3) Robustness to Tag Quality: To explicitly quantify how much the weakly paired tags contribute to the final generation, we conduct a dedicated inference-time ablation study by injecting noise into the tags before the Interpretation Phase. As shown in Table VI, completely removing the tags ("Without Tags") results in a catastrophic performance drop (S_m^* falls to 58.95), proving that the tags are the major driver and indispensable semantic anchors of our framework. When we randomly drop 60% or 80% of the tags to simulate

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Fig. 4. Qualitative visualization of the WeCap framework on the LEVIR-CC and WHU-CDC datasets. Each example showcases our two-stage inference pipeline. The model first executes the *Perception Phase* to generate a set of semantically meaningful tags that act as an interpretable "semantic anchor." The subsequent *Interpretation Phase* weaves these tags into a fluent caption. Our framework adeptly handles diverse scenarios across both datasets, including complex new developments (e.g., "building with a parking lot"), object removal ("a road has disappeared"), and robustly identifies no-change scenes.

missed detections, the performance degrades smoothly rather than collapsing entirely, showcasing the LMM's robustness to incomplete information. Interestingly, when a false tag is deliberately added ("Add 1 Noise Tag"), the score drops significantly to 35.32. This phenomenon counterintuitively highlights a strength: the model remains highly faithful to the provided visual tags rather than stubbornly hallucinating changes based purely on language priors.

4) Analysis of Fine-Grained Perception: A critical assumption of our factorized paradigm is $p(C|T, P) \approx p(C|T)$. To empirically investigate whether this abstraction limits the model's ability to capture fine-grained visual details, we evaluated the recall of fine-grained vocabulary on the test set, as detailed in Table VII. For quantity/extent words, WeCap achieves a recall of 25.60%, which is on par with the fully paired Qwen2-VL-7B SFT (25.43%) but slightly lower than the traditional SEN model (30.70%). However, for spatial/directional words, WeCap achieves a recall of 21.65%, significantly outperforming both the fully paired SEN (15.70%) and the fully paired Qwen2-VL-7B SFT (14.28%). This compelling result confirms that our model does not merely hallucinate spatial details based on language priors. On the contrary, while end-to-end fully paired LMMs tend to suffer from mode collapse and ignore spatial specifics, WeCap's semantic tags successfully act as structural anchors, triggering the LMM's robust pretrained visual grounding capabilities to perceive locations from the image pair.

5) Vocabulary Bias and General Capabilities: To ensure that our weakly supervised alignment does not overly restrict the LMM's vocabulary or cause catastrophic forgetting of its broad knowledge, we evaluate the pre- and post-finetuned models on standard multimodal benchmarks. As shown in Table VIII, WeCap-2B maintains comparable performance to the base Qwen2-VL-2B model across benchmarks like MMMU, MME, and MMStar. This indicates that our Linguistic Alignment strategy effectively aligns the output style with RSICC domain experts without imposing harmful vocabulary bias or diminishing the model's generalization capabilities for unseen concepts.

F. Qualitative Analysis

To provide an intuitive understanding of our model's behavior and its real-world effectiveness, we present a series of qualitative examples from both the LEVIR-CC and WHU-CDC datasets in Fig. 4. These visualizations highlight

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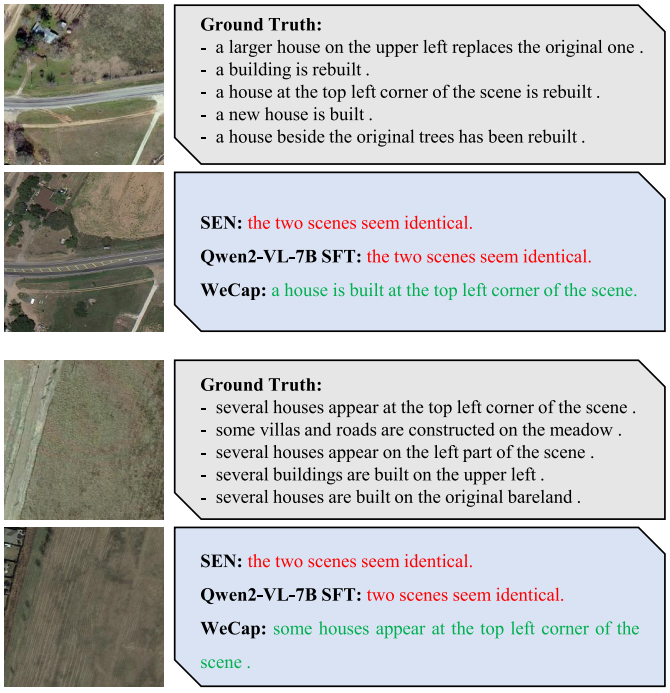


Fig. 5. Side-by-side qualitative comparison of captions generated by the SEN, Qwen2-VL-7B, and weakly paired WeCap.

the interpretability, robustness, and generalization capabilities of our WeCap framework.

1) Interpretability of the Two-Stage Pipeline: The figure clearly illustrates the step-by-step reasoning process enabled by our factorized design. In every case, the *Perception Phase* first distills the visual changes into a set of discrete, human-readable semantic tags. For instance, in the bottom-left example from LEVIR-CC, the model correctly identifies "[villas, bareland]" as the involved objects and "[built]" as the action. These tags serve as a transparent intermediate representation, making the model's "thought process" explicit before the *Interpretation Phase* synthesizes them into a fluent sentence: "villas are built on the bareland in the top left."

2) Versatility and Robustness: The selected examples showcase the model's versatility in handling a wide range of change scenarios. It successfully describes complex, multiobject changes, such as the appearance of a "building with a parking lot" on LEVIR-CC. Critically, the model is not limited to describing additions; it can also identify and articulate the removal of objects, as demonstrated in the WHU-CDC example where it correctly perceives "[road, bareland]" and "[disappeared]." Furthermore, the framework exhibits strong robustness by consistently and correctly identifying multiple no-change scenarios on both datasets, generating "[None]" tags and an appropriate description. This is a crucial capability for practical applications, as it minimizes false positives.

3) Side-by-Side Comparison with Baselines: To better highlight the strengths of our proposed framework, Fig. 5 presents a qualitative side-by-side comparison between WeCap, SEN, and Qwen2-VL-7B SFT. As observed, traditional models and even fully paired LMMs frequently struggle to generalize on complex scenes, erroneously generating "the two scenes seem identical" despite obvious changes. In stark contrast, WeCap accurately detects and describes visual changes, demonstrating superior semantic accuracy and robustness against false negatives.

V. Conclusion

In this work, we introduced WeCap, a weakly paired framework that dismantles the reliance on expensive, fully aligned data in RSICC. By factorizing the captioning process through intermediate semantic tags and training a single LMM via a unified multitask objective, WeCap achieves remarkable data efficiency. Our extensive experiments demonstrate that our approach not only rivals or surpasses fully supervised SOTA methods but also exhibits superior cross-dataset generalization and the unique ability for unilateral data scaling. We believe this factorized, prompt-driven paradigm offers a promising path toward more scalable, robust, and interpretable vision-language models in specialized domains.

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